TRPO (Trust Region Policy Opt.)

- A2C: $\pi(a|s,\theta)$
- $J(\theta) = \mathbb{E}_{p(\tau|\theta)} R(\tau) \to \max_\theta$

$$\nabla J(\theta) = \frac{1}{1-\gamma} \mathbb{E}_{s\sim\mu^\pi(s)} \mathbb{E}_{a\sim\pi} \nabla_\theta \log \pi(a|s,\theta) A^\pi(s,a)$$

$$L(\theta) = -\mathbb{E}_{s\sim\mu^\pi(s)} \mathbb{E}_{a\sim\pi} \log \pi(a|s,\theta) A^\pi(s,a)$$

θ, $\pi(a|s,\theta)$ — current policy.

$\tilde{\theta}$, $\tilde{\pi}(a|s,\tilde{\theta})$ — new policy.

Th $J(\tilde{\theta}) = J(\theta) + \mathbb{E}_{p(\tau|\tilde{\theta})} \sum_t \gamma^t A^{\pi_\theta}(s_t, a_t)$.

$$\mathbb{E}_{p(\tau|\tilde{\theta})} \sum_t \gamma^t A^{\pi}(s_t, a_t) = \mathbb{E}_{p(\tau|\tilde{\theta})} \sum_t \gamma^t \left(Q^{\pi}(s_t, a_t) - V^{\pi}(s_t) \right) =$$

$$= \mathbb{E}_{p(\tau|\tilde{\theta})} \sum_t \gamma^t \left(r_t + \gamma V^{\pi}(s_{t+1}) - V^{\pi}(s_t) \right) =$$

$$= \mathbb{E}_{p(\tau|\tilde{\theta})} \sum_{t=0}^{+\infty} \left(\gamma^t r_t + \gamma^{t+1} V^{\pi}(s_{t+1}) - \gamma^t V^{\pi}(s_t) \right) = \tilde{J}(\tilde{\theta}) - \overbrace{\mathbb{E}_{p(s_0)} V^{\pi}(s_0)}^{J(\theta)}$$

$$J(\tilde{\theta}) - J(\theta) = \mathbb{E}_{p(\tau|\tilde{\theta})} \sum_t \gamma^t A^{\pi}(s_t, a_t) =$$

$$= \left\{ \text{in style of PG main theorem} \right\} =$$

$$= \frac{1}{1-\gamma} \mathbb{E}_{s \sim \mu^{\tilde{\pi}}(s)} \sum_a \tilde{\pi}(a|s,\tilde{\theta}) A^{\pi}(s,a)$$

$$\mu^{\tilde{\pi}}(s) = \sum_t \gamma^t p(s_t = s | t, \tilde{\pi})$$

surrogate function

$$L_{\theta}(\tilde{\theta}) := J(\theta) + \frac{1}{1-\gamma} \mathbb{E}_{s \sim \mu^{\tilde{\pi}}(s)} \sum_a \tilde{\pi}(a|s,\tilde{\theta}) A^{\pi}(s,a)$$

$$L_\theta(\tilde{\theta})\Big|_{\tilde{\theta}=\theta} = J(\theta) + \frac{1}{1-\gamma}\mathbb{E}_{s\sim\mu^\pi(s)}\underbrace{\sum_a \pi(a|s,\theta)A^\pi(s,a)}_{=0}$$

$$\ominus J(\theta)$$

$$\mathbb{E}_\pi\left(Q^\pi(s,a) - V^\pi(s)\right) = V^\pi(s) - V^\pi(s) = 0$$

$$\nabla_\theta L_\theta(\tilde{\theta})\Big|_{\tilde{\theta}=\theta} = \frac{1}{1-\gamma}\mathbb{E}_{s\sim\mu^\pi(s)}\sum_a \nabla_{\tilde{\theta}}\pi(a|s,\tilde{\theta})\Big|_{\tilde{\theta}=\theta} A^\pi(s,a) = \{\text{PG main theorem}\} =$$

$$= \nabla J(\theta)$$

$$TV(p\|q) := \frac{1}{2}\int |p(x)-q(x)|\,dx$$

$$\alpha := TV^{\max}(\theta\|\tilde{\theta}) = \max_s TV(\pi(\cdot|s,\theta)\|\tilde{\pi}(\cdot|s,\tilde{\theta}))$$

- **Th** $J(\tilde{\theta}) \geq L_\theta(\tilde{\theta}) - \frac{4\varepsilon\gamma}{(1-\gamma)^2}\alpha^2$, $\varepsilon = \max_{a,s}|A^\pi(s,a)|$

$$KL(p\|q) \geq 2TV^2(p\|q)$$

$$J(\tilde{\theta}) \geq L_\theta(\tilde{\theta}) - \frac{2\varepsilon\gamma}{(1-\gamma)^2} KL^{\max}(\theta\|\tilde{\theta}) \to \max_{\tilde{\theta}}$$

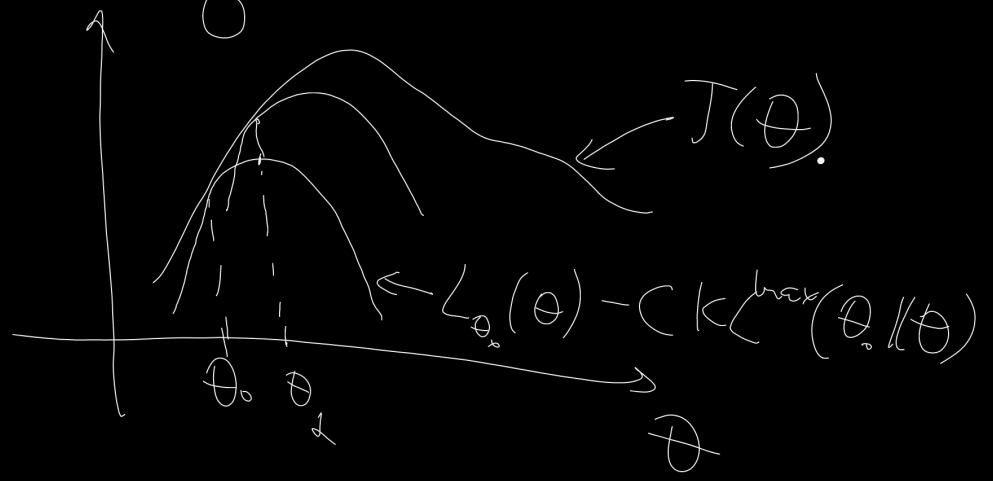
$\theta_0,\ J(\theta_0)$

$$J(\theta_1) \geq L_{\theta_0}(\theta_1) - C D_{KL}^{max}(\theta_0 \| \theta_1) \geq$$

$$\geq L_{\theta_0}(\theta_0) - C D_{KL}^{max}(\theta_0 \| \theta_0) = J(\theta_0)$$

$$\| \qquad\qquad \|$$

$$J(\theta_0) \qquad 0$$



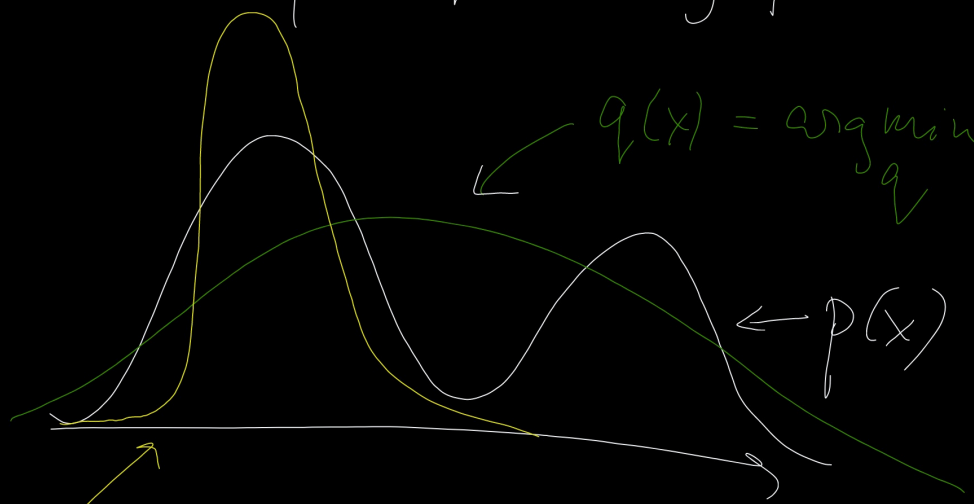
$$L_\theta(\tilde{\theta}) = J(\theta) + \frac{1}{1-\gamma} \mathbb{E}_{s\sim\mu^\pi(s)} \mathbb{E}_{a\sim\pi} \frac{\tilde{\pi}(a|s,\tilde{\theta})}{\pi(a|s,\theta)} A^\pi(s,a)$$

$$L_\theta(\tilde{\theta}) - C KL^{\max}(\theta \| \tilde{\theta}) \to \max_{\tilde{\theta}}$$

$$\begin{cases} L_\theta(\tilde{\theta}) \to \max_{\tilde{\theta}} \\ KL^{\max}(\theta \| \tilde{\theta}) \le \delta \end{cases} \Rightarrow \begin{cases} L_\theta(\tilde{\theta}) \to \max_{\tilde{\theta}} \\ \hat{KL}(\theta \| \tilde{\theta}) \le \delta \end{cases}$$

$$\hat{KL}(\theta \| \tilde{\theta}) = \mathbb{E}_{s\sim\mu^\pi(s)} KL(\pi(\cdot|s,\theta) \| \tilde{\pi}(\cdot|s,\tilde{\theta}))$$

$$KL(p \,||\, q) = \int p(x) \log \frac{p(x)}{q(x)} \, dx$$



$$q(x) = \arg\min_q KL(q \,||\, p)$$

$$KL(\theta \| \tilde{\theta}) \approx \overset{0}{KL(\theta \| \theta)} + \nabla_{\tilde{\theta}} KL(\theta \| \tilde{\theta})\Big|_{\tilde{\theta}=\theta}^{\overset{0}{}} (\tilde{\theta}-\theta) +$$

$$+ \frac{1}{2}(\tilde{\theta}-\theta)^T \underbrace{\nabla^2_{\tilde{\theta}} KL(\theta \| \tilde{\theta})\Big|_{\tilde{\theta}=\theta}}_{H} (\tilde{\theta}-\theta)$$

$$H \in \mathbb{R}^{\dim\Theta \times \dim\Theta}$$

$$H = \mathbb{E}_{s,a} \nabla_\theta \log \pi(a|s,\theta) \nabla \log \pi^T$$

$H^T = H \geq 0$ – Fisher Information Matrix

$$\begin{cases} L_\theta(\theta) + \underbrace{\nabla_{\tilde\theta} L_\theta(\tilde\theta)\Big|_{\tilde\theta=\theta}}_{g}^{T}(\tilde\theta-\theta) \to \max\limits_{\tilde\theta} \\ \frac{1}{2}(\tilde\theta-\theta)^T H(\tilde\theta-\theta) \le \delta \end{cases}$$

$$Lagrange(\tilde\theta, \lambda) = -g^T(\tilde\theta-\theta) + \lambda\left(\frac{1}{2}(\tilde\theta-\theta)^T H(\tilde\theta-\theta) - \delta\right)$$

KKT: $\nabla_\theta Lagrange(\tilde\theta, \lambda) = -g + \lambda H(\tilde\theta-\theta) = 0 \Rightarrow$

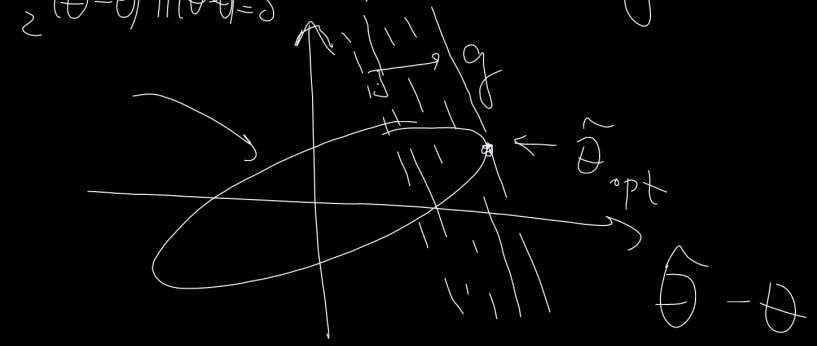
$$\Rightarrow \tilde\theta = \theta + \frac{1}{\lambda} H^{-1} g$$

$$\lambda \geq 0, \quad \lambda\left(\frac{1}{2}(\tilde{\theta}-\theta)^T H(\tilde{\theta}-\theta)-\delta\right)=0$$

$\lambda = 0$ – impossible

$$\frac{1}{2}(\tilde{\theta}-\theta)^T H(\tilde{\theta}-\theta)=\delta \Rightarrow \frac{1}{2}\frac{1}{\lambda}g^T H^{-1} H \frac{1}{\lambda} H^{-1} g =$$

$$= \frac{1}{2\lambda^2} g^T H^{-1} g = \delta \Rightarrow \lambda^2 = \frac{g^T H^{-1} g}{2\delta}$$



$$d = H^{-1} g \iff Hd = g$$

- $d = CG(H\{\cdot\}, g, \text{num_iters})$

  $H\{v\} = Hv$

HVP in PyTorch

$f(x)$, Find: $\nabla^2 f(x) \cdot v$

$\nabla f(x)$, $h(x) := \nabla f(x)^T v$

$\nabla h(x) = \nabla^2 f(x) v$

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grad_f = torch.autograd.grad(f(x), x, create_graph=True)
grad_f_v = torch.sum(grad_f * v)
hvp = torch.autograd.grad(grad_f_v, x)
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$$d \approx H^{-1} g$$

$$\tilde{\theta} = \theta + \beta d$$

$$\frac{1}{2}(\tilde{\theta} - \theta)^T H (\tilde{\theta} - \theta) = \delta \Rightarrow \beta = \sqrt{\frac{2\delta}{d^T H d}}$$

Line Search: While True:

$$\tilde{\theta} = \theta + \beta d$$

if $L_\theta(\tilde{\theta}) \geq L_\theta(\theta)$ and $\hat{KL}(\theta \| \tilde{\theta}) \leq \delta$, then break

$\beta = \beta \cdot \alpha$, $\alpha < 1$

TRPO scheme

- Init. $\theta_\pi, \theta_V, \delta = 0.01$

  For iterations:

  $B := \emptyset$ ← rollout length = 1024, 2048,

  For each parallel rollout:

  Collect $s_0, a_0, r_0, \dots, s_{n-1}, a_{n-1}, r_{n-1}, s_n$ using $\pi(a|s, \theta_\pi)$

  $GAE_n = 0$

  For $t = n-1, n-2, \dots, 0$:

  $\delta_t = r_t + \gamma V(s_{t+1}|\theta_V) - V(s_t|\theta_V)$

  $GAE_t = \delta_t + (\lambda\gamma)\, GAE_{t+1}$

  $y_t = GAE_t + V(s_t|\theta_V)$

  $\pi_t = \pi(a_t|s_t, \theta_\pi)$

  $\mu_t = \mu(s_t, \theta_\pi)$

  $\sigma_t = \sigma(s_t, \theta_\pi)$

  Add $\{s_t, a_t, GAE_t, y_t, \pi_t, \mu_t, \sigma_t\}$ to B

$$GAE = Normalize(GAE)$$

$$\theta_T = Adapt\ Policy(B)$$

$$\theta_V = Adapt\ Value(B)$$

Adapt Value (B)

For epochs:

For batch b in B:

$$L = \frac{1}{|b|} \sum_{t \in b} \frac{1}{2} (V(s_t|\theta) - y_t)^2$$

$$\theta = \text{opt_step}(\nabla_\theta L)$$

Adapt Policy (B)

$L := 0$, $KL := 0$

For $t = 0, 1, \ldots, |B|$:

$$L = L + \frac{\pi(a_t | s_t, \theta)}{\pi_t} GAE_t$$

$$KL = KL + \frac{1}{2} \log \frac{\sigma(s_t, \theta)}{\sigma_t} - \frac{1}{2} + \frac{1}{2\sigma^2(s_t, \theta)} \left(\sigma_t^2 + (\mu_t - \mu(s_t, \theta))^2 \right)$$

- $g = \nabla_\theta L$, $H(v) = \nabla^2 KL \cdot v + \rho v$, $\rho = 0.01$ – hyperparam
- $d = CG(H(\cdot), g, num_iters = 10)$
- $\beta = \sqrt{\frac{2\delta}{d^T H d}}$

while True:

$$\tilde{\theta} = \theta + \beta d$$

$$L = \sum_{t \in B} \frac{\pi(a_t | s_t, \tilde{\theta})}{\pi_t} \cdot GAE_t$$

$$KL = \sum_{t \in B} KL(\pi(\cdot | s_t, \theta) \,\|\, \pi(\cdot | s_t, \tilde{\theta}))$$

if $L \geq 0$ and $KL \leq \delta$, then break

$\beta = \beta \cdot \alpha$

$$\begin{cases} g^T(\tilde{\theta} - \theta) \to \max\limits_{\tilde{\theta}} \\ \frac{1}{2}\|\tilde{\theta} - \theta\|^2 \le \delta \end{cases} \Rightarrow \tilde{\theta} = \theta + \beta \cdot g$$

$$g = \nabla L_\theta(\tilde{\theta})\Big|_{\tilde{\theta}=\theta} = \nabla J(\theta)$$

TRPO with $H = I$ $=$ A2C